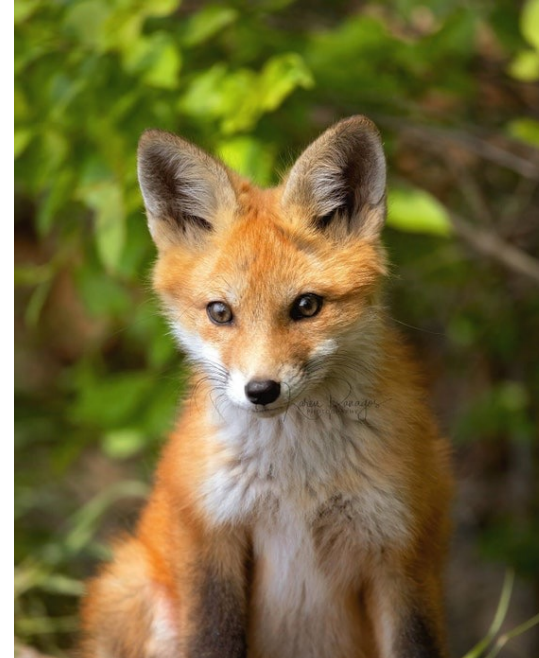


Machine Learning

Dr. Ahmed Aljarah

Machine learning (ML)

- Learning is the process by which a system improves its performance through experience.
- It is a branch of Artificial Intelligence that focuses on creating algorithms that help computers change and improve their behavior using data.
- How we learn?



Machine learning

Definition by Tom Mitchell (1998):

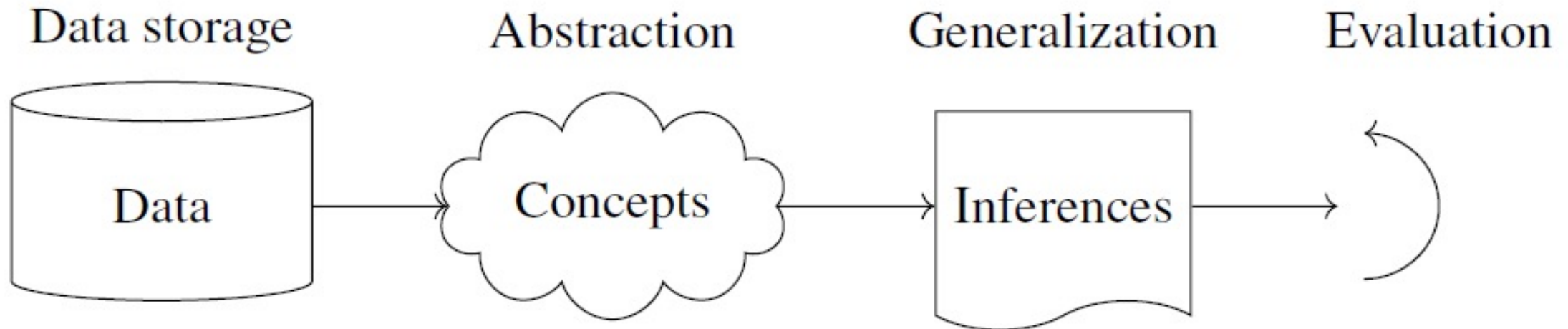
A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks T , as measured by P , improves with experience E .

Example: A robot driving learning problem

- Task T : Driving on highways using vision sensors.
- Performance measure P : Average distance traveled before an error.
- Training experience E : it learns by watching pictures of driving and the actions a human took.

Components of the machine learning process

- Data Storage: Stores and retrieves large amounts of data (humans use the brain, computers use memory).
- Abstraction: Turns raw data into useful information by building models.
- Generalization: Uses learned knowledge to solve new, similar problems.
- Evaluation: Checks performance and improves using feedback.



ML computations

Machine Learning (ML) is about teaching computers to improve their performance using data and past experience.

Learning means adjusting a model's parameters using training data.

A model can make predictions, discover patterns in data, or do both.

Data in ML

- Machine learning process starts with collecting and preparing data.
- If the data is poor, the results will also be poor. A model is only as good as the data it learns from.
- Data preparation may seem simple, but it is the most important step, since it can determine success or failure.
- In ML, there are different types of data, especially:
 - Labeled data
 - Unlabeled data

Labeled data

- Labeled data is data where each example has a known correct answer (label).
- The label shows what the output or class should be and is usually given by experts or a reliable process.

Example:

In an image classification task, labeled data would consist of images along with their corresponding class labels (ex. cat, dog, car).

Unlabeled data

- Unlabeled data is a dataset where the instances or examples do not have any associated labels or target variables.
- The data consists only of input features, without any corresponding output or classification.

Example:

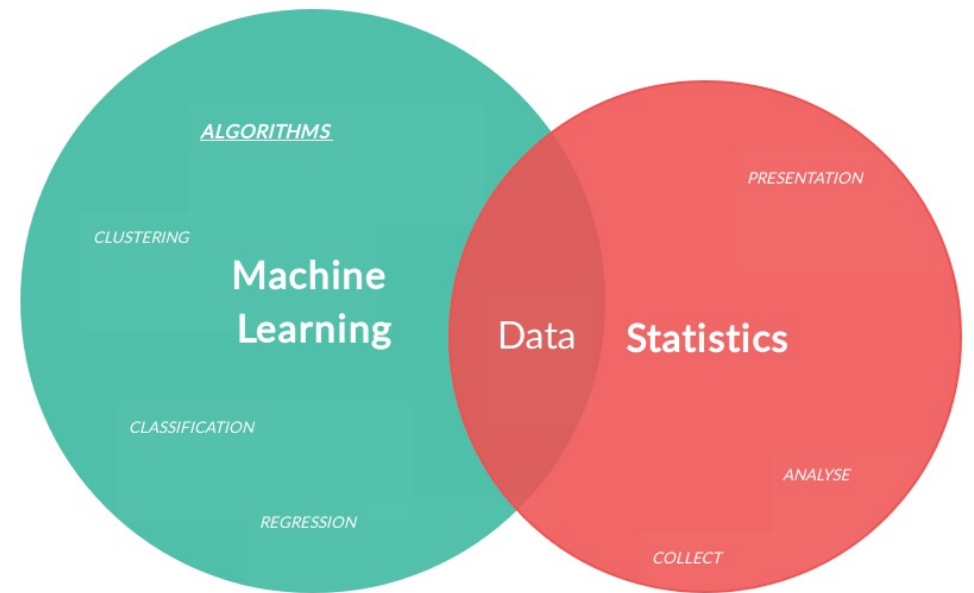
A collection of images without any labels or annotations.

Main types of data used in training

- Structured data: refers to data that is organized in a clear format, usually in tables with rows and columns.
 - Tabular data: Data in spreadsheets, databases, or CSV files, where rows are examples and columns are features.
 - Time-series data: Data collected over time (stock prices, sensor readings, or weather data).
- Unstructured data: refers to data that does not follow a fixed format, such as text, images, audio, and video.
- It usually needs more advanced ML techniques to find useful patterns.

Relation between ML and Statistics

- Statistics is the branch of mathematics that allows us to analyze and gain knowledge about populations from a sample.
- Machine Learning uses statistical theory to build mathematical models, since its main task is to make inferences from a sample.



Statistics & ML

- Statistics helps us understand a large group (population) by studying a smaller part of it (sample).

For example, instead of asking everyone in a city about their income, we study a small group and use it to make conclusions about the whole city.

- Machine Learning uses ideas from statistics to build models that learn patterns from data. These models can then make predictions or decisions on new data.

In simple terms:

ML = statistics + algorithms to learn from data and make predictions.

The role of the computer scientist

The role of the computer scientist in Machine Learning is twofold:

- Learning (training): Efficient algorithms are used to learn a model by processing large amounts of data and solving optimization problems.
- Inference (production): After training, the model must make predictions efficiently. Its speed and memory usage are just as important as its accuracy.

Learning

Learning is done through a set of examples that are given as input (training) to the algorithm so it can learn.

Each example is made up of a set of input data, and, in some cases, the exact output(s) obtained from those inputs.

Format: $(inputs, [output])$ or $(i_1, i_2, \dots, i_n, s)$

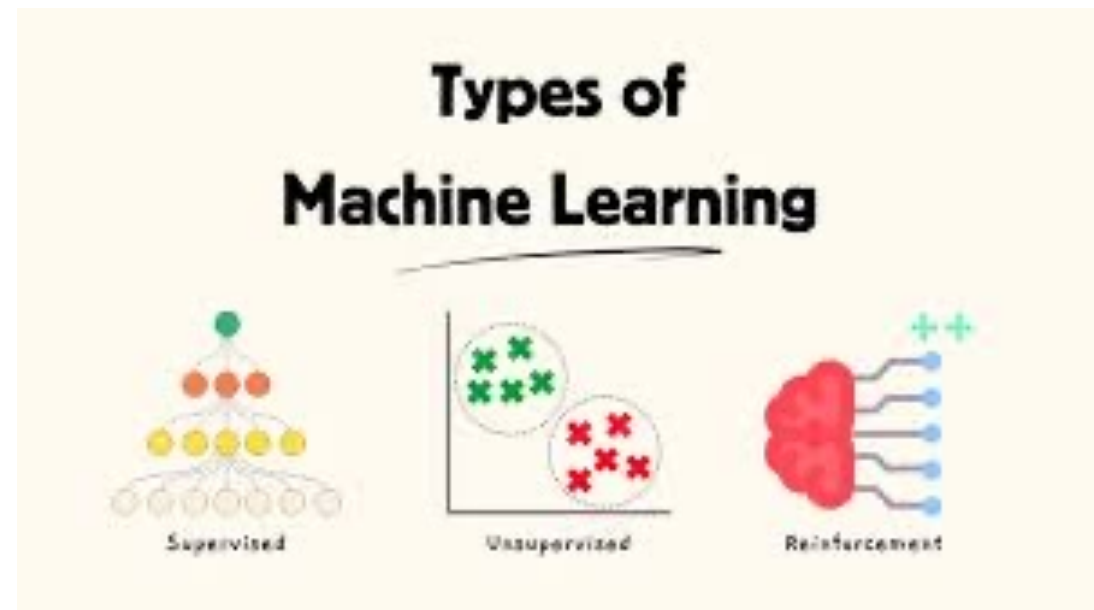
ML Algorithms Classification

Machine learning algorithms are grouped into three main types, based on how they learn from data:

Supervised Learning

Unsupervised Learning

Reinforcement Learning



Supervised learning

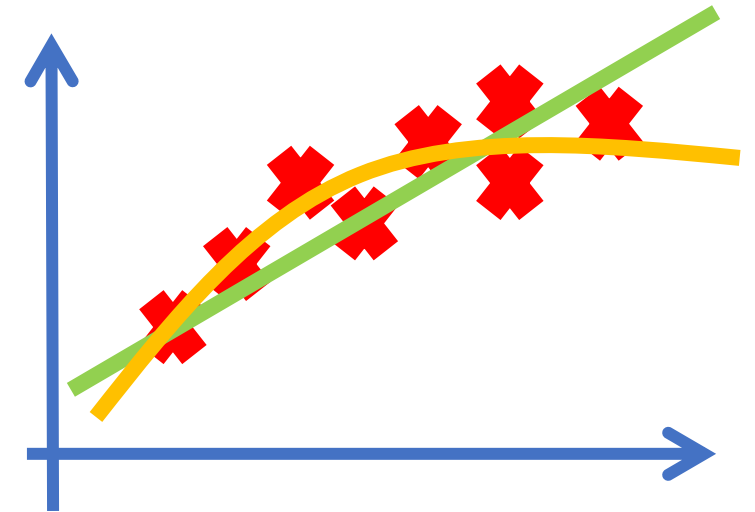
The model learns from examples that already have the correct answers (labels).

It compares its predictions with the real answers and adjusts itself to reduce errors.

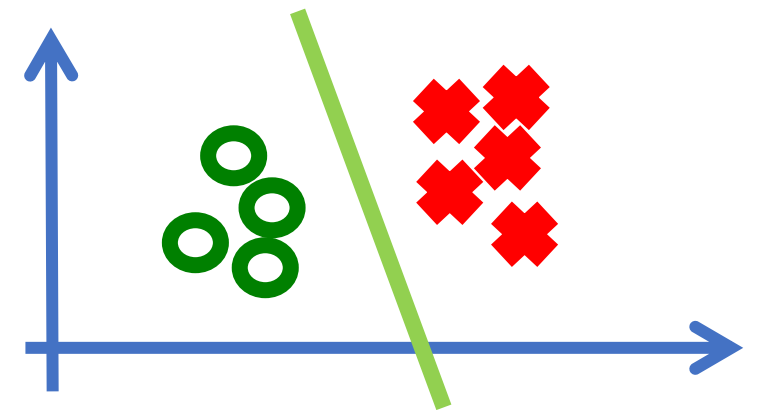
1. Prediction: predicting numbers (e.g., house prices, temperatures).
2. Classification: choosing a category (e.g., spam vs. not spam, cat vs. dog).

Example:

Training a model with emails labeled as spam or not spam so it learns to classify new emails.



Prediction or forecasting



Classification

Training Data

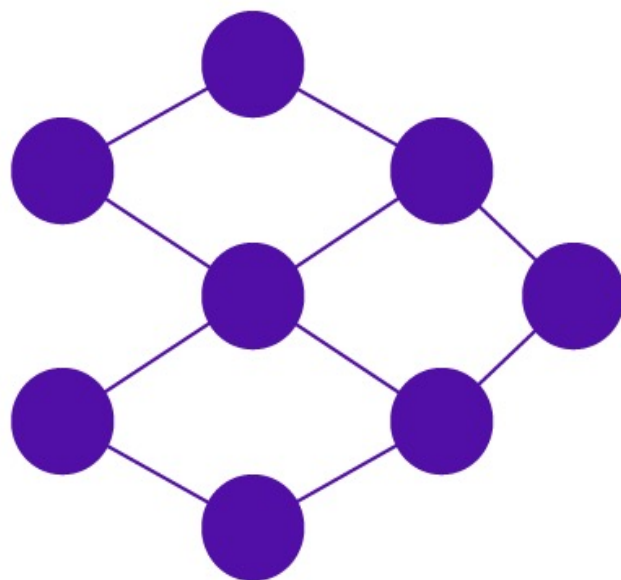
ML Algorithm

Model

Prediction

Apple

Banana



Class: Banana



**Unseen and
unlabeled data**

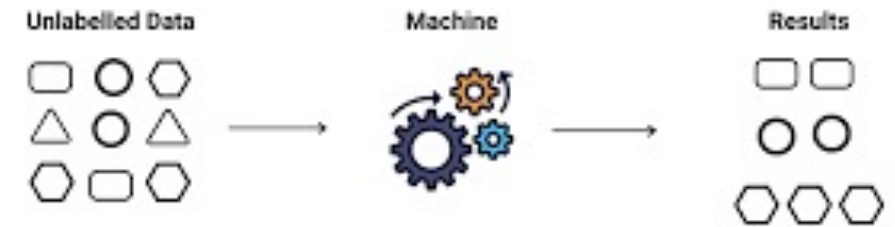
Unsupervised learning

- The model learns from data without labels.
 - There are no correct answers given, so the model looks for patterns, structures, or groups by itself.
- Clustering: grouping similar data points.
 - Association/Description: discovering hidden patterns or relationships in data.

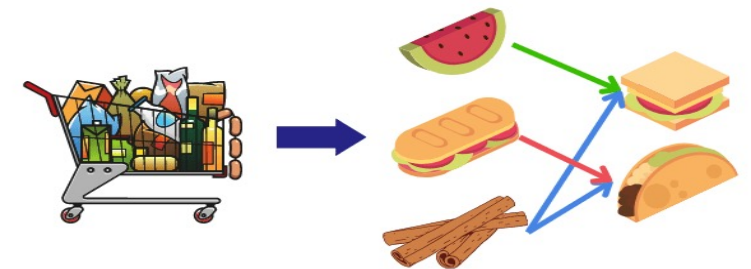
Example:

Grouping customers by behavior without knowing the groups in advance.

Unsupervised Learning

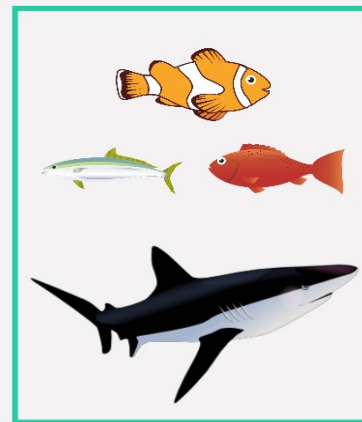
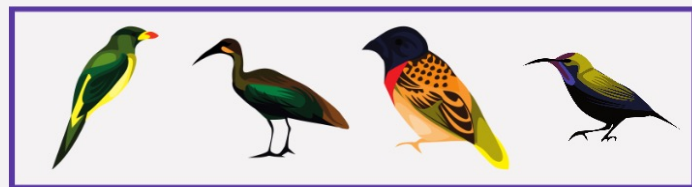
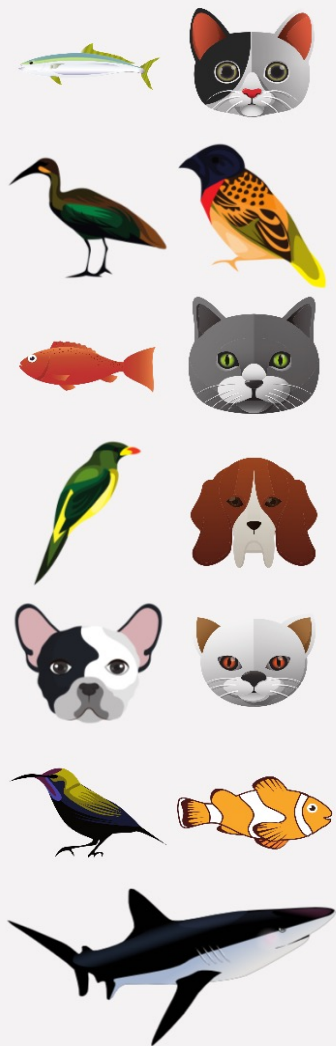


Clustering



Association

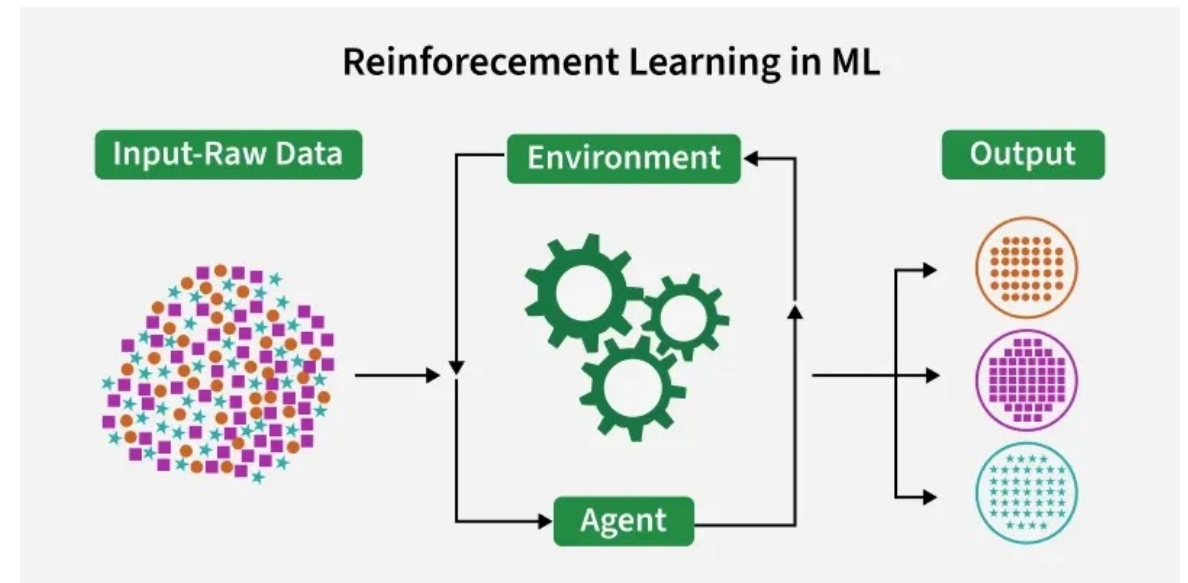
No labels



Reinforcement learning

- The model learns by trial and error through interaction with an environment.
- It does not get correct answers, but instead receives:
 1. Positive reward for good actions
 2. Negative reward for bad actions
- Over time, it learns the best actions to maximize rewards.

Example: A robot learning to walk, or an AI learning to play a game by winning or losing.



Feature selection

- The process of choosing only the important input variables (features) for a model.
- Removes irrelevant or noisy data.
- Keeps only the features that improve model performance.
- Does not change the data, only selects or removes features.



Example: Car broke down

Model	Brand	Owner's name	year	KM	color
Civic	Honda	Carlos	2016	85,000	White
Corolla	Toyota	Sara	2012	120,000	Black
Altima	Nissan	María	2010	95,000	Red
Elantra	Hyundai	Jose	2019	40,000	Blue

feature
selection



Model	Brand	year	KM
Civic	Honda	2016	85,000
Corolla	Toyota	2012	120,000
Altima	Nissan	2010	95,000
Elantra	Hyundai	2019	40,000

Feature selection advantages



Reduces noise in the data
Prevent learning from noise



Makes the model simpler and faster
Reduce training time



Improves accuracy and performance



Reduces overfitting

Understanding ML as function learning

- In Machine Learning, the goal is to learn a function that maps inputs to outputs (input $\rightarrow$ output). Sometimes, this function gives probabilities instead of exact values.
- We use ML because this function is usually unknown or too complex to write manually. Real-world problems are hard to describe with fixed rules.
- If we already knew the function, we could just program it directly:
input $\rightarrow$ function $\rightarrow$ output, and no learning would be needed

Example:

Problem: Predict house prices.

Input: house size, location, number of rooms

Output: price

There is a hidden function like: $\text{price} = f(\text{size}, \text{location}, \text{rooms})$

Machine learning fundamentals

- Building a machine learning model involves data collection and preparation.
- Selecting an appropriate algorithm
- Training the model on the prepared data
- Evaluating its performance.



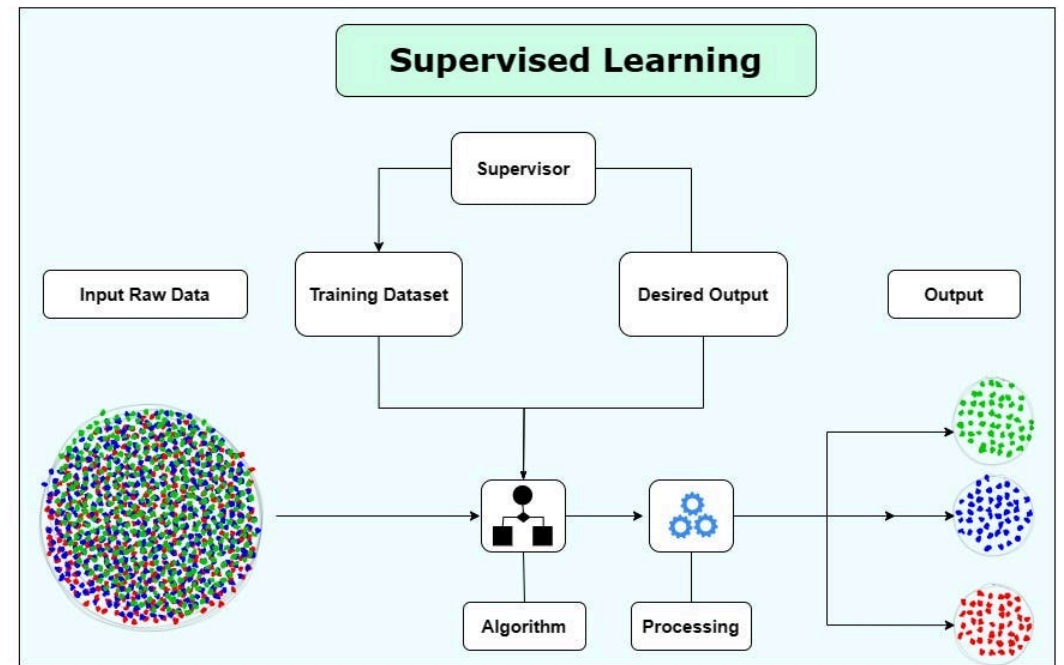
Inferencing

- After a model is trained, it is used to make predictions or decisions.
- There are two main types of inferencing:
 1. Batch inferencing: Processes large amounts of data all at once. It focuses more on accuracy than speed (e.g., data analysis).
 2. Real-time inferencing: Processes data instantly as it arrives. It focuses on speed and quick decisions (e.g., chatbots, self-driving cars).



Supervised ML

- It is a learning approach in which each training example is represented as an ordered pair consisting of an input and its corresponding output.
- The goal is to learn a function that maps inputs to outputs while minimizing prediction error.
- The output of each training example represents the desired result and serves as the label for that example.



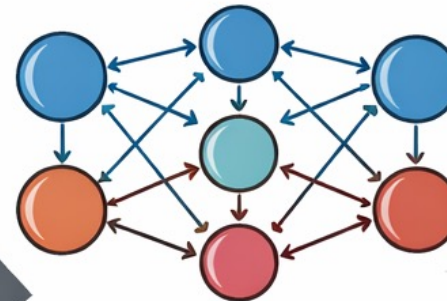
- The main objective is to learn a function that maps inputs to outputs using labeled training data.
- Supervised machine learning techniques are commonly divided into two main categories according to the type of function that the model aims to learn:
 1. **Regression (or prediction)** focuses on learning a function that produces a continuous output, allowing the model to estimate or predict a numerical value for a given input.
 2. **Classification**, in contrast, involves learning a function with a discrete output, where the objective is to assign a given input to one class within a finite set of possible classes

Inductive Learning (Example-Based Learning)

A learning approach where each training example is represented as an ordered pair (input, output).

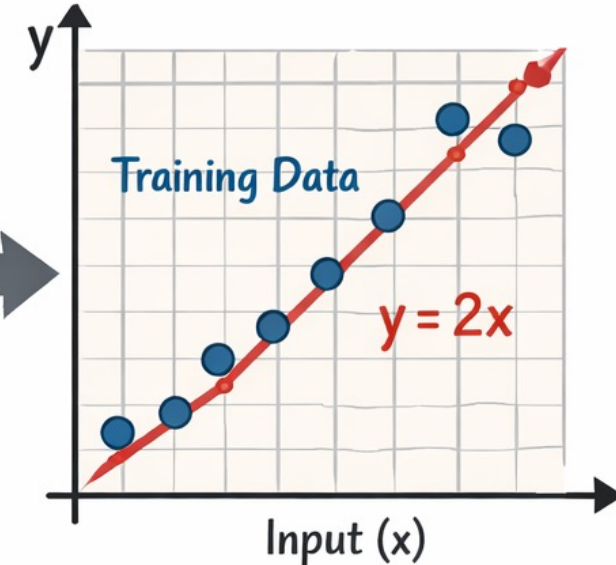
Training Examples	
Input	Output (Label)
(2, 4)	→ 4
(3, 6)	→ 6
(4, 8)	→ 8
(5, 10)	→ 10

Learn
Function



$$y = 2x$$

Best Fit Model



✓ Minimise Prediction Error

✓ Output = Label

10

Learn the Function that Maps Input to Output.

Supervised machine learning techniques

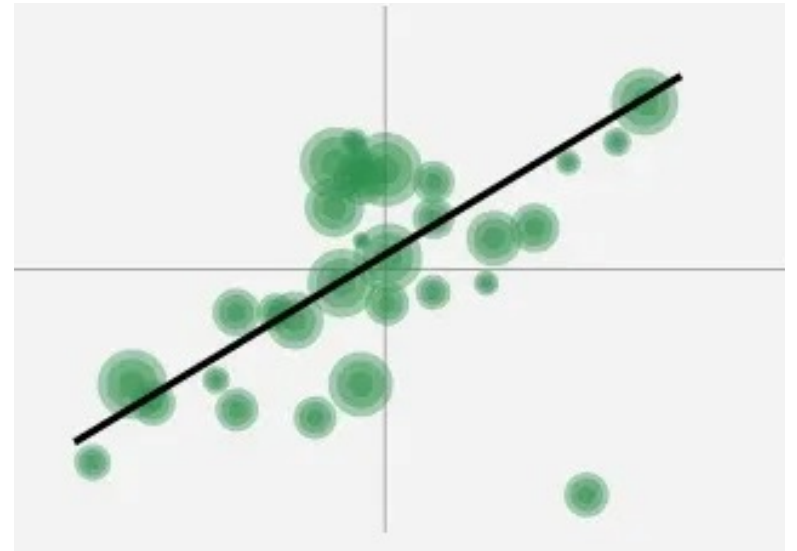
There are many techniques for performing supervised machine learning.

Covered ones are:

- Linear regression
 - Simple (single variable)
 - Multiple (multivariable)
- Logistic regression
- Decision trees
- Random forest
- Neural networks

Linear regression

- The term regression originates from the concept of regression towards the mean.
- Although the modern meaning of regression no longer directly reflects the original idea of returning to the mean, the term has remained in use.
- Linear regression is an algorithm that learns from labeled data to model the relationship between input and output using a linear function.
- It assumes a straight-line relationship between input and output, where the output changes at a constant rate with respect to the input.



Understanding linear regression

Linear Regression is a statistical model used to describe and predict the relationship between an independent variable (input) and a dependent variable (output).

It examines two main aspects:

- Which variables are significant predictors of the outcome variable.
- How well the regression line can make accurate predictions.

Simple linear regression



A statistical technique used to analyze and model relationships between variables.



Applied in many fields, including engineering.



Considered one of the most widely used statistical methods.

Simple linear regression

Regression Problem: Given a set of data points (x, y) , where x is the input and y is the output, the objective is to find a function f such that

$$y = f(x) \text{ that best fits the data}$$

A common source of confusion is that the function f may be nonlinear with respect to the variable x , while still being linear with respect to its parameters.

Example:

House price prediction

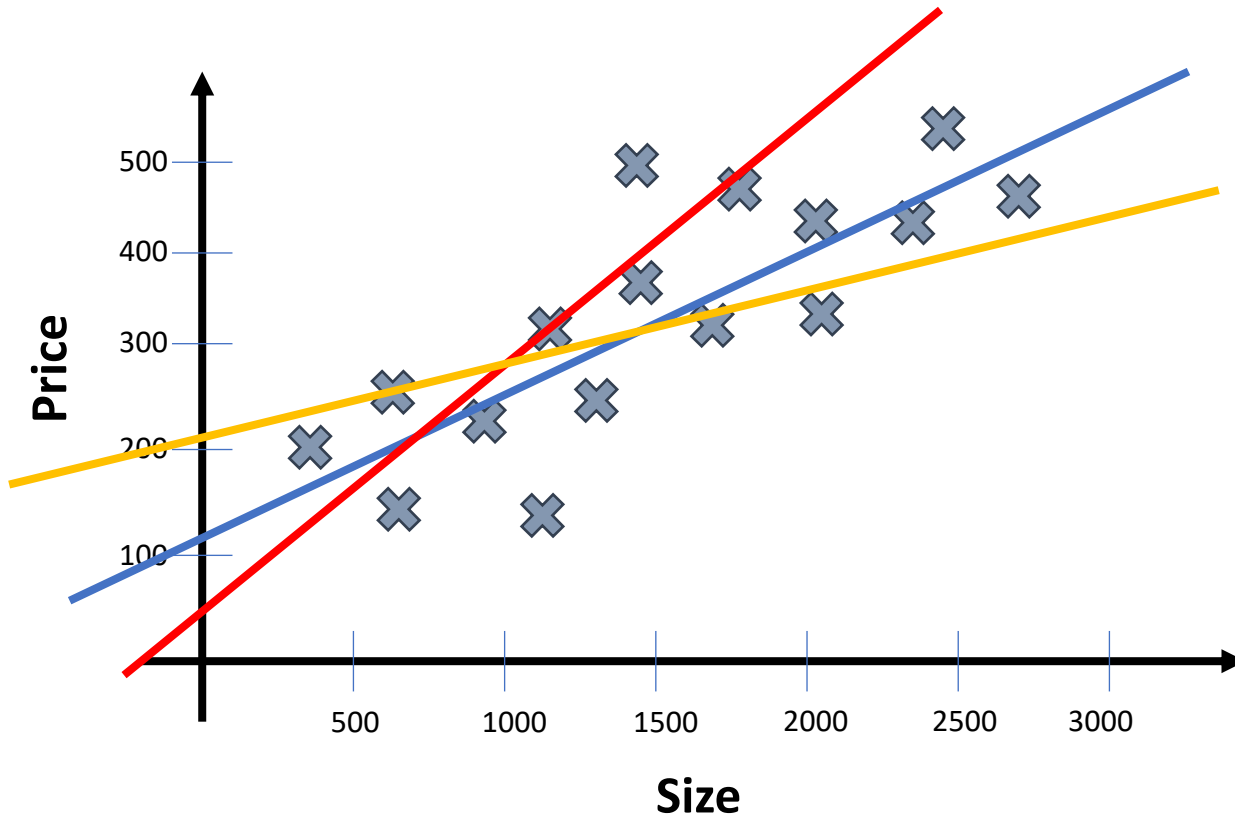
The objective is to determine the relationship between the selling price of a house and its size.

Suppose we have the following data:

X	Y
Size (ft ²)	Price (USD 1000's)
2104	460
1416	232
1534	315
852	178
...	...

Training set

Graphical description



1. What is the function that best fits the data points?
2. The function is not necessarily a straight line

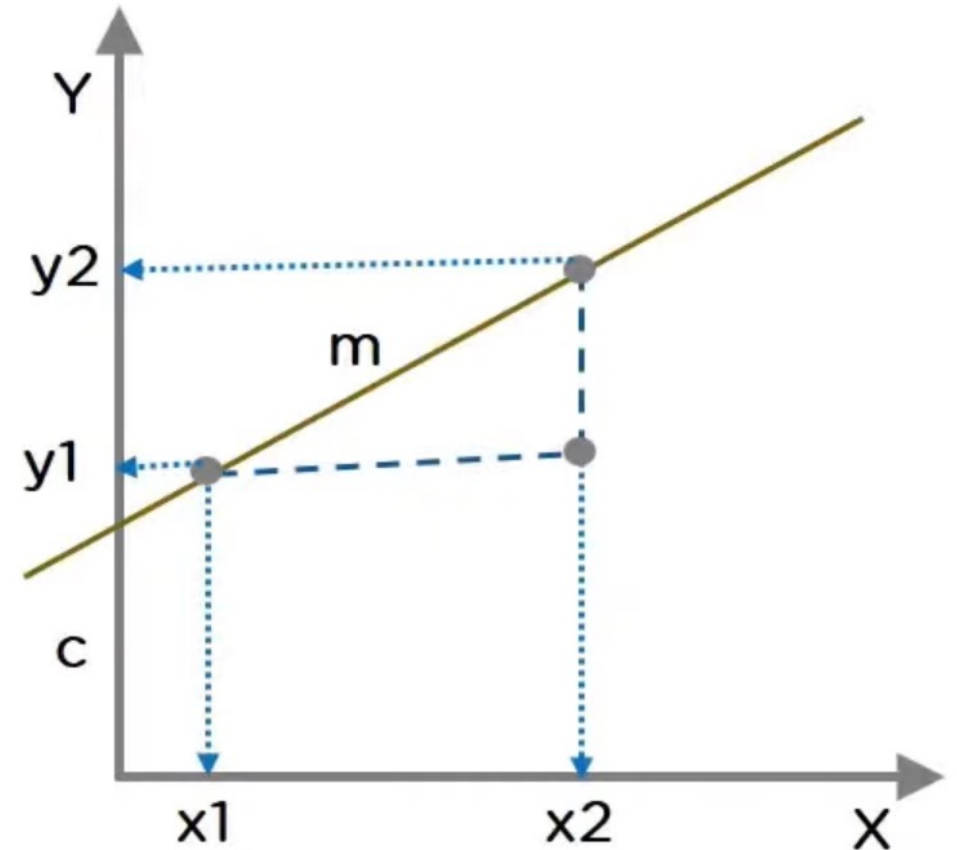
Equation of simple linear regression

$$y = mx + c$$

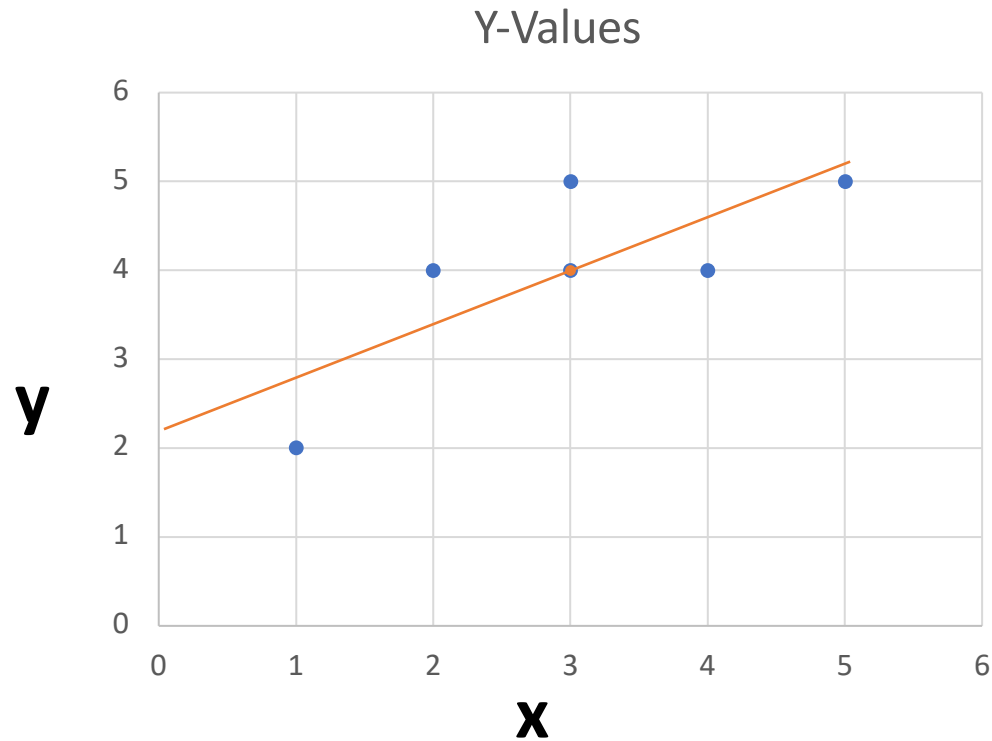
Where:

- y is the predicted value (dependent variable)
- x is the input (independent variable)
- m is the slope of the line (how much y changes when x changes)
- c is the intercept (the value of y when $x = 0$)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



Math behind the linear regression



$$M = \frac{(n * \sum(x * y)) - (\sum(x) * \sum(y))}{(n * \sum(x^2)) - (\sum(x))^2}$$

$$c = \frac{(\sum(y) * \sum(x^2)) - (\sum(x) * \sum(x * y))}{(n * \sum(x^2)) - (\sum(x))^2}$$

Independent variable dependent variable

x	y
1	2
2	4
3	5
4	4
5	5

Mean 3 4

x	y	x²	y²	x*y
1	2	1	4	2
2	4	4	16	8
3	5	9	25	15
4	4	16	16	16
5	5	25	25	25

$$\sum 15$$

$$\sum 20$$

$$\sum 55$$

$$\sum 86$$

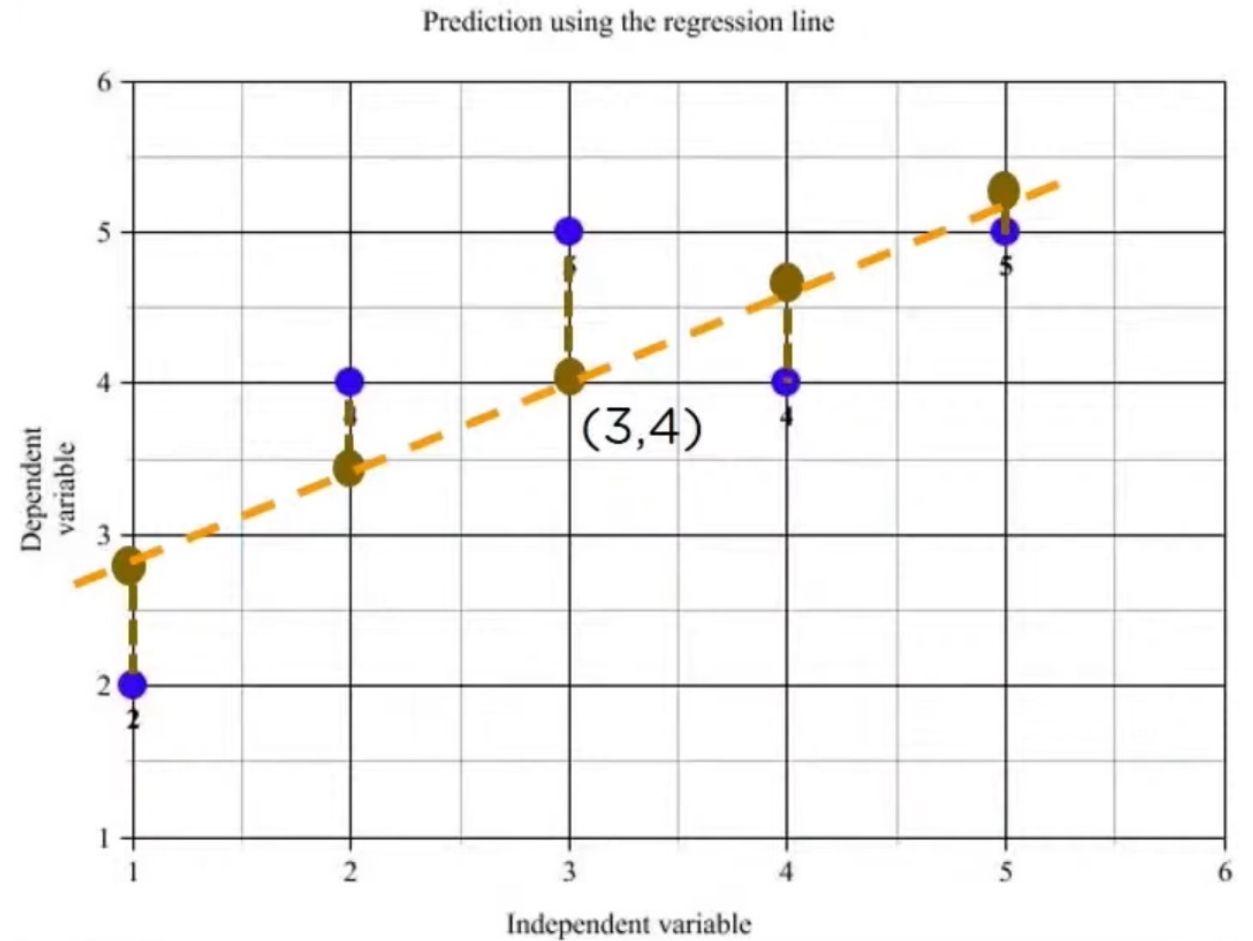
$$\sum 66$$

$$M = \frac{(n * \sum(x*y)) - (\sum(x) * \sum(y))}{(n * \sum(x^2)) - (\sum(x))^2} = \frac{(5*66) - (15*20)}{(5*55) - (255)} = 0.6$$

$$c = \frac{(\sum(y) * \sum(x^2)) - (\sum(x) * \sum(x*y))}{(n * \sum(x^2)) - (\sum(x))^2} = \frac{(20*55) - (15*66)}{(5*55) - (255)} = 2.2$$

$$y = xm + c = 3*0.6 + 2.2 = 4$$

x	y	y _{predict}
1	2	$y = 1*0.6+2.2 = 2.8$
2	4	$y = 2*0.6+2.2 = 3.4$
3	5	$y = 3*0.6+2.2 = 4$
4	4	$y = 4*0.6+2.2 = 4.6$
5	5	$y = 5*0.6+2.2 = 5.2$



Multiple linear regression

Simple Linear Regression

$$y = mx + c$$

OR

$$y = \theta_0 + \theta_1 x$$

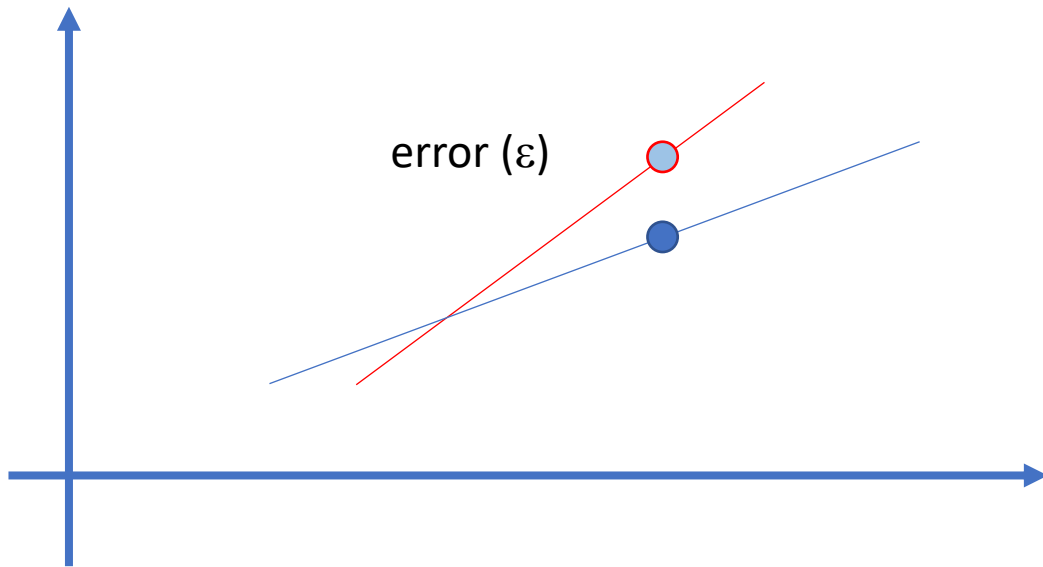
Multiple Linear Regression

$$y = m_1 * x_1 + m_2 * x_2 + \dots + m_n * x_n + c$$

OR

$$y = \theta_0 + \theta_1 * x_1 + \theta_2 * x_2 + \dots + \theta_n * x_n$$

Prediction errors



- It is the distance between the actual dependent data point and the point predicted by the model.
- The challenge is to select the model that minimizes the error across all observations simultaneously.
- How can all observations be included in the calculation?

Least Square Error

- The most common method is the least squares method, which defines an error function based on all data points.
- The best model is obtained by minimizing this function.
- One approach is to choose the line that minimizes the total error for all observations.

x	y	y_{predict}	$y - y_{\text{predict}}$	$(y - y_{\text{predict}})^2$
1	2	2.8	- 0.8	0.64
2	4	3.4	0.6	0.36
3	5	4	1	1
4	4	4.6	-0.6	0.36
5	5	5.2	- 0.2	0.04

$$\sum 2.4$$

Mean Square Error (MSE)

MSE is an evaluation metric that calculates the average of the squared differences between the actual and predicted values for all the data points.

It is a way to quantify the accuracy of a model's predictions.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - y_{\text{predict}_i})^2 = \frac{2.4}{5} = 0.48$$

x	y	y _{predict}	y - y _{predict}	(y - y _{predict}) ²
1	2	2.8	- 0.8	0.64
2	4	3.4	0.6	0.36
3	5	4	1	1
4	4	4.6	-0.6	0.36
5	5	5.2	- 0.2	0.04

$$\sum 2.4$$

Coefficient of Determination (R-squared)

- **R-squared (R^2)** is a statistical measure that indicates how much of the variation in the data is explained by the model.
- Its value ranges from **0 to 1**. In general, a higher R^2 value means that the model fits the data better.
- Mathematically, it can be expressed as:

$$R^2 = 1 - \left(\frac{RSS}{TSS} \right)$$

- **Residual Sum of Squares (RSS)**: The Residual Sum of Squares is the sum of the squared residuals for all data points. It measures the difference between the observed values and the values predicted by the model.

$$RSS = \sum (y_i - y_{i_predict})^2$$

- **Total Sum of Squares (TSS)**: The Total Sum of Squares is the sum of the squared differences between each observed value and the mean of the target variable. It measures the total variation in the data.

$$TSS = \sum (y_i - \bar{y})^2$$

Where:

y_i = actual values (y_test)

$y_{i_predict}$ = predicted values ($y_predict$)

$\bar{y}$ = mean of the actual values

Calculate the R-square

x	y	y _{predict}	y - y _{predict}	(y - y _{predict}) ²	(y _i - $\bar{y}_i$) ²
1	2	2.8	- 0.8	0.64	4
2	4	3.4	0.6	0.36	0
3	5	4	1	1	1
4	4	4.6	-0.6	0.36	0
5	5	5.2	- 0.2	0.04	1

R-square evaluation:

- $R^2 \approx 0.9 - 1.0 \rightarrow$ Excellent fit
- $R^2 \approx 0.7 - 0.9 \rightarrow$ Good fit
- $R^2 \approx 0.5 - 0.7 \rightarrow$ Moderate / acceptable
- $R^2 < 0.5 \rightarrow$ Weak model

$$RSS = \sum (y_i - y_{i_predict})^2 = 2.4$$

$$TSS = \sum (y_i - \bar{y}_i)^2 = 6$$

$$R^2 = 1 - \left(\frac{RSS}{TSS} \right) = 1 - \frac{2.4}{6} = 0.6.$$

This means that the model explains about 60% of the variation in the observed data, while the remaining 40% of the variation is not explained by the model.

Machine learning

Machine learning process



Define the task.

- What do you want the model to do?

Select the model.

- What do you want the model to do?

Collect the data.

- Clean the data
- Split into validation set (if necessary)

Train the model.

- Choose model type
- Set some parameters

Evaluate the model.

- Choose evaluation metrics
- Assess performance on validation set (if necessary)

Example 1:

Implement a linear regression model using python

1000_Companies

R&D Spend	Administration	Marketing Spend	State	Profit
165349.2	136897.8	471784.1	New York	192261.83
162597.7	151377.59	443898.53	California	191792.06
153441.51	101145.55	407934.54	Florida	191050.39
144372.41	118671.85	383199.62	New York	182901.99
142107.34	91391.77	366168.42	Florida	166187.94
131876.9	99814.71	362861.36	New York	156991.12
134615.46	147198.87	127716.82	California	156122.51
130298.13	145530.06	323876.68	Florida	155752.6

...

```
# Data Visualization Building the  
#Correlation matrix
```

```
sns.heatmap(df.select_dtypes(include=  
'number').corr())
```

```
# Importing the libraries  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
# Connect colab with dirve  
from google.colab import drive  
drive.mount('/content/drive')
```

```
#Importing the dataset  
df = pd.read_csv("/content/drive/  
MyDrive/AI  
course/1000_Companies.csv")  
df.head()  
df.shape  
df.info()  
df.size
```



```
# Encoding categorical data
from pandas.core.indexes import category
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
#le = LabelEncoder()
#df.iloc[:,3] = le.fit_transform(df.iloc[:,3])
#print(df)
#Extracting the Independent and Dependent variables
x = df.iloc[:, :-1]
y = df.iloc[:, 4]
ohe = OneHotEncoder( sparse_output=False).set_output(transform='pandas')
en = ohe.fit_transform(df[['State']])
x = x.join(en)
x = x.drop(columns=['State'])
#x = pd.concat([x,en], axis = 1).drop(columns=["State"])
print(x)
```

```
# Splitting the dataset into the Training set and Test set
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test= train_test_split(x,y,test_size=0.2,random_state=0)
```

```
# Fitting Multiple Linear Regression to the Training set
from sklearn.linear_model import LinearRegression
regressor =LinearRegression()
regressor.fit(x_train,y_train)
```

```
# Predicting the Test set results
```

```
y_predict = regressor.predict(x_test)
```

```
Print(y_predict )
```

```
# Calculating the Coefficients
```

```
print(regressor.coef_)
```

```
# Calculating the Intercept
```

```
print(regressor.intercept_)
```

```
# Calculating the R squared value
```

```
from sklearn.metrics import r2_score
```

```
r2_score(y_test, y_predict)
```

```
# Create new dataset and apply the model on it
```

```
data = { 'R&D Spend':[165349.20,142107.34 ], 'Administration':[136897.800,91391.770],
```

```
'Marketing Spend': [471784.1000,366168.4200],
```

```
'State_California':[0.0,0.0], 'State_Florida':[0.0,1.0],
```

```
'State_New York':[1.0,0.0]}
```

```
x1 = pd.DataFrame(data)
```

```
x1.head()
```

```
y1 = regressor.predict(x1)
```

```
print(y1)
```

Activity 2.2:

Linear regression

- Use the dataset `data_salary.csv` to design and implement a complete linear regression model.
- The activity should include data exploration, preprocessing, model development, evaluation, and prediction tasks.
 - Analyze and interpret relationships between input variables
 - Prepare and preprocess data for machine learning
 - Develop a linear regression model
 - Evaluate the model and interpret results
 - Predict salaries for the final 500 records

